WEEK 3 – SOC ANALYST INTERNSHIP REPORT

Linux Logs, Endpoint Telemetry, Auditing and Security Monitoring

Internship week	Week 3
Reporting date	6 November 2026
Focus areas	SOC alert investigation, logging, audit trails, network detection concepts

Prepared as a weekly learning and activity report.

This report records the learning activities, practical work discussed, prepared documentation, and tasks that remain skipped or require evidence. It intentionally does not claim completion of activities that were not verified.

1. Executive Summary

Week 3 focused on understanding endpoint and Linux security telemetry, reviewing logs, learning SIEM alert investigation, and studying network intrusion detection concepts. The learner worked through Ubuntu auditd installation and rule-based command auditing, and reviewed the resulting audit record. Six LetsDefend lesson writeups and supporting reference documents were prepared. Windows Sysmon and Snort practical installation/testing were skipped during this work period.

2. Learning Objectives

• Understand how endpoint telemetry supports SOC investigations.

• Learn the purpose of Sysmon and important Windows event types.

• Understand Linux auditd, audit rules, and audit event fields.

• Learn how Snort rules identify selected network traffic patterns.

• Understand SIEM alert triage, log analysis, and incident-management workflows.

• Relate selected detection activity to MITRE ATT&CK techniques, including T1046.

3. Learning Activities

SIEM alert investigation: Reviewed a structured workflow: review the alert, understand the detection, examine supporting logs, establish context, correlate evidence, assess the alert, respond or escalate, and document the result.

Security solutions: Prepared study material covering layered security controls such as firewalls, IDS/IPS, endpoint protection, EDR, SIEM, email/web security, identity controls, and vulnerability management.

Network log analysis: Prepared study material about firewall, DNS, proxy/web, and flow records, including key fields and the importance of correlating network activity with endpoint and identity evidence.

SIEM fundamentals: Prepared study material about log sources, collection, parsing and normalization, indexing, correlation, dashboards, searches, alerts, and operational considerations.

Incident management: Prepared study material about incident identification, ticket creation, prioritization, ownership, investigation, escalation, containment, remediation, and closure.

Linux incident response: Prepared study material about authentication logs, system journal, processes, network connections, users and privileges, persistence checks, and evidence preservation.

4. Hands-on Work

4.1 Ubuntu auditd

The Ubuntu VM was used to install and start auditd. A rules file was created under /etc/audit/rules.d/ to monitor changes to /etc/passwd, /etc/shadow, and /etc/group, and to record execve system calls. The rules were loaded and checked with auditctl. The learner searched audit records using ausearch.

A sample audit record shared during the work showed:

Field	Observed value	Interpretation
comm	ausearch	Command name recorded by auditd
exe	/usr/sbin/ausearch	Executable path
auid	rajdeep	Original login identity
uid	root	Effective user identity for this command
success	yes	System call completed successfully
key	command_execution	Audit rule key that matched

The sample confirms that auditd recorded execution of the ausearch command. It is evidence of audit logging, but it is not evidence of a malicious event. The learner also reviewed common fields such as type, msg, exe, uid, gid, auid, comm, and key.

4.2 Windows Sysmon

The Sysmon installation and event-generation task was skipped. Therefore, this report does not claim that Sysmon was installed or that Event IDs 1, 3, or 11 were captured. The prepared reference materials describe these event types for later study and practical completion.

4.3 Snort

The learner's Ubuntu VM reported Ubuntu 26.04.1 LTS (Resolute). The command `sudo apt install snort -y` returned “Unable to locate package snort” after `apt update`. Snort installation and rule testing were then skipped. No Snort alerts or successful installation are claimed in this report.

5. Detection Concepts and Reference Materials

• Windows Security events: 4624 (successful logon), 4625 (failed logon), 4688 (process creation), and 4720 (user account created).

• Sysmon events: 1 (process creation), 3 (network connection), 11 (file creation), and 22 (DNS query).

• Linux auditd record types include EXECVE, USER_LOGIN, and USER_AUTH.

• Snort starter examples cover ICMP ping, SSH connection attempts, HTTP traffic, and a threshold-based TCP SYN scan detection rule.

• The Nmap SYN scan example is mapped to MITRE ATT&CK T1046, Network Service Discovery.

6. Prepared Documentation

• Separate theory notes for the Week 3 questions.

• Event ID and rule cheat sheet with more than 15 reference entries.

• Snort rule cheat sheet containing starter examples and the T1046 example.

• Separate TryHackMe room writeups/templates.

• Separate LetsDefend lesson writeups for all six listed lessons.

• Week 3 README with folder structure, activity status, and submission checklist.

7. Activity Status and Evidence Gaps

Activity	Status recorded	Evidence / next action
Ubuntu auditd	Worked through	Keep screenshots of service status, loaded rules, and ausearch output.
Windows Sysmon	Skipped	Do not claim installation or event capture; complete later only if time permits.
Snort	Skipped after package lookup failure	No installation or alert evidence; consider a supported OS/version later.
TryHackMe rooms	Writeups prepared; completion not verified here	Complete each room and add actual screenshots and answers.
LetsDefend	Six writeups prepared; Lesson 1 completion was confirmed	Verify each remaining lesson in the account and add genuine completion screenshots.
PDF / presentation / submission	In progress	Review, add evidence, and upload final folder to internship Drive.

8. Key Learnings

• SIEM alerts are starting points for investigation; analysts must validate them with supporting evidence and context.

• Correlating endpoint, authentication, and network logs helps create a more complete incident timeline.

• Linux auditd can record selected system activity based on configured rules and supports searches with ausearch.

• Audit fields such as auid, uid, exe, comm, success, and key help explain who performed an action and which rule recorded it.

• Detection rules should be tested in an authorized lab and tuned to reduce false positives.

• A clear report must distinguish completed practical work from prepared study material and skipped tasks.

9. Conclusion

Week 3 strengthened foundational understanding of SOC monitoring, SIEM alert investigation, Linux auditing, network-log analysis, and incident-management concepts. The auditd work provided a practical example of collecting and interpreting Linux audit records. Sysmon and Snort practical tasks remain incomplete, and the learning-room completion evidence must be attached before final submission. The final internship folder should contain only genuine screenshots, verified results, and documentation that accurately reflects the work performed.